MTH501 - Linear Algebra

(Mid-Term) 2021 GRAND Quiz 90 TO 100% Correct Solve (Alhamdulillah)

BY Virtual Assistance Sonu Ilyas Mughal

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https://www.youtube.com/channel/UCjboEX65C_yN7as5di1OuIQ/playlists Sadiq Ahmed

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MTH501 - Linear Algebra

MTH501:Grand Quiz

Quiz Start Time: 12:27 PM, 30 June 2

Question # 1 of 30 (Start time: 12:27:51 PM, 30 June 2021)

Total Mark

If a system of equations is solved using the Jacobi's method , then which of the following is the most appropriate answer about the matrix M that is derived from the coefficient matrix ?

Select the correct option

- All of its entries below the diagonal must be zero.
- All of its entries below and above the diagonal must be zero.
- All of its entries above the diagonal must be zero.
- All of its entries on the diagonal must be zero.

Click to Save Answer & Move to Next Questio

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Question # 2 of 30 (Start time: 12:28:22 PM, 30 June 2021)

If A and B are matrices then A+B is only possible if both matrices have same -------

Select the correct option

- (●) Order
- ( ) Rows
- ( ) Columns
- ( ) Elements

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Question # 3 of 30 (Start time: 12:28:45 PM, 30 June 2021)

The Gauss-Seidel method is applicable to symmetric ----------definite matrices

Select the correct option

- Zero
- Positive
- Equal
- Negative

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Question # 4 of 30 (Start time: 12:29:14 PM, 30 June 2021)

The transpose of the product of two matrices is equal to

Select the correct option

- ○ $(AB)^T = A^T B^T$
- ● $(AB)^T = B^T A^T$

Clic

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Question # 5 of 30 (Start time: 12:31:23 PM, 30 June 2021)

The system of equations :

$x_1 = 1,$

$x_2 = 0;$ can be expressed in the form - - - - - .

Select the correct option

- ○ $Ax = 1$ $By = 0$
- ○ $Ax = 0$
- ● $Ax = b$
- ○ $Ax = 0$ $By = 1$

Click to Save

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Quiz Start Time: 12:27 PM, 30 June 2021

Question # 6 of 30 (Start time: 12:31:47 PM, 30 June 2021)

Total Marks: 1

$$x_1 + 2x_2 + 3x_3 = 7$$
$$4x_1 + x_2 + 2x_3 = 2$$
$$-4x_1 + 3x_2 + 9x_3 = 4$$

The augmented matrix for the system is

Select the correct option

Reload Math Equations

- ○ $\begin{bmatrix} 1 & 2 & 3 & 7 \\ -4 & 1 & 2 & 2 \\ 4 & 3 & 9 & 4 \end{bmatrix}$
- ● $\begin{bmatrix} 1 & 2 & 3 & 7 \\ 4 & 1 & 2 & 2 \\ -4 & 3 & 9 & 4 \end{bmatrix}$
- ○ $\begin{bmatrix} 1 & 2 & 3 \\ -4 & 1 & 2 \\ 4 & 3 & 9 \end{bmatrix}$
- $\begin{pmatrix} 1 & 2 & 3 \end{pmatrix}$

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Question # 7 of 30 (Start time: 12:32:15 PM, 30 June 2021)

The Gaussian elimination procedure is one of several method to solve the

Select the correct option

- Procedure matrix
- Eliminated matrix
- Determinant matrix
- Inverse of matrix

Click t

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Quiz Start Time: 1

Question # 8 of 30 (Start time: 12:32:45 PM, 30 June 2021)

The Sytem of equations : $y = 0$ and $x = 5$ has - - - - solution(s).

Select the correct option

Re

- Infinite many
- distinct multiple
- No
- Unique

Click to Save Answer & Mov

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Question # 9 of 30 (Start time: 12:33:14 PM, 30 June 2021)

The three most important special cases of R^n are $x \in R$

Select the correct option

- R, R^2, R^3
- R, N

Click

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Question # 10 of 30 (Start time: 12:33:48 PM, 30 June 2021)

A decomposition of a matrix as a product of two or more matrices is called the matrix _____.

Select the correct option

- transformation
- composition
- factorization
- multiplication

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Quiz S

Question # 11 of 30 (Start time: 12:34:19 PM, 30 June 2021)

If B is the matrix obtained by interchanging two rows or columns of the square matrix A, then det (B) = _____.

Select the correct option

- -1/det(A)
- ● - det(A)
- det(A)
- 1/det(A)

Click to Save Ans

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Quiz Start T

Question # 12 of 30 (Start time: 12:34:48 PM, 30 June 2021)

In LU-decomposition, if A is the square matrix of order m, then the order of both lower and upper triangular matrices will be _____.

Select the correct option

- m^2
- m
- $m+1$
- $2m$

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Question # 13 of 30 (Start time: 12:35:34 PM, 30 June 2021)

The determinant of a triangular matrix is the sum of the entries of the main diagonal.

Select the correct option

- False
- True

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Question # 14 of 30 (Start time: 12:36:02 PM, 30 June 2021)

The Invertible Matrix Theorem applies only to _____ matrices.

Select the correct option

- square
- identity
- rectangular
- scalar

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Quiz Start Time: 12:27 PM

Question # 15 of 30 (Start time: 12:36:24 PM, 30 June 2021)

$(AB)^t = ?$

Select the correct option

Reload M

- BA
- $B^t A^t$

Click to Save Answer & Move to N

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Quiz St

Question # 16 of 30 (Start time: 12:36:59 PM, 30 June 2021)

Which of the following is NOT the axiom for vector space where u, v, w in V are set of vectors and l, m, n are scalars?

Select the correct option

- ○ (l +m) u= l u+ m u
- ● u.v =v.u
- ○ l (u + v)= l u + l v
- ○ u + (v + w) = (u + v) + w

Click to Save Answ

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Question # 17 of 30 (Start time: 12:37:22 PM, 30 June 2021)

An m x m _____ matrix is only square matrix that is both unit lower triangular and upper triangular.

Select the correct option

- null
- diagonal
- identity
- scalar

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Question # 18 of 30 (Start time: 12:37:48 PM, 30 June 2021)

A determinant does not change if we add a multiple of a row to another row.

Select the correct option

- FALSE
- TRUE

Click t

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Question # 19 of 30 (Start time: 12:38:12 PM, 30 June 2021)

LU-factorization is a matrix decomposition which writes a matrix as the product of a _____ matrix and an upper triangular matrix.

Select the correct option

- lower triangular
- identity
- diagonal
- null

Click to Save

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Quiz Start Time: 12:27 PM, 3

Question # 20 of 30 (Start time: 12:38:35 PM, 30 June 2021)

T

Which of the following will be the Linear Combinition corresponding to $\begin{pmatrix} -2 & 3 \\ 5 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$?

Select the correct option

Reload Math

- $\begin{pmatrix} -2 \\ 3 \end{pmatrix} x + \begin{pmatrix} 5 \\ 1 \end{pmatrix} y$
- $\begin{pmatrix} 5 \\ 1 \end{pmatrix} x + \begin{pmatrix} -2 \\ 3 \end{pmatrix} y$
- $\begin{pmatrix} 3 \\ 1 \end{pmatrix} x + \begin{pmatrix} -2 \\ 5 \end{pmatrix} y$
- (●) $\begin{pmatrix} -2 \\ 5 \end{pmatrix} x + \begin{pmatrix} 3 \\ 1 \end{pmatrix} y$

Click to Save Answer & Move to Next

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Quiz Start Time: 12:27 PM,

Question # 21 of 30 (Start time: 12:39:05 PM, 30 June 2021)

The equation : $0x - 0y = -5$ has - - - - solution(s).

Select the correct option

Reload Mat

- (●) No
- distinct finite
- Infinite many
- Unique

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Quiz Start Time: 12:27 PM,

Question # 21 of 30 (Start time: 12:39:05 PM, 30 June 2021)

The equation : $0x - 0y = -5$ has - - - - solution(s).

Select the correct option

Reload Mat

- No
- distinct finite
- Infinite many
- Unique

Click to Save Answer & Move to Nex

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Quiz Start Tim

Question # 22 of 30 (Start time: 12:40:03 PM, 30 June 2021)

Which of the following system will have the non - trivial solution?

Select the correct option

- $x = y = 0$
- $x = y = 1$

Click to Save Answer & M

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Question # 23 of 30 (Start time: 12:40:34 PM, 30 June 2021)

The determinant of a diagonal matrix is the product of the diagonal elements.

Select the correct option

- (●) TRUE
- ( ) FALSE

Cli

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Quiz Start Tim

Question # 24 of 30 (Start time: 12:40:55 PM, 30 June 2021)

Which of the following would be the value of t, if $\begin{bmatrix} 2e^{-3t} \\ 5e^{-3t} \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \end{bmatrix}$?

Select the correct option

- $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$
- zero
- $\frac{\begin{bmatrix} 2 \\ 5 \end{bmatrix}}{\begin{bmatrix} 2 \\ 5 \end{bmatrix}} = 1$
- $\pm\frac{1}{3}$

Click to Save Answer & N

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Quiz Start

Question # 25 of 30 (Start time: 12:41:19 PM, 30 June 2021)

For the following system of Equations:

$x + 2y = 0,\ 2x - y = 0;$

the solution set is - - - - - .

Select the correct option

- ○ Line through origin in $\mathbb{R}^3$
- ○ Any arbitrary line in $\mathbb{R}^2$
- ○ Real line $\mathbb{R}$
- ● Line through origin in $\mathbb{R}^2$

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Question # 26 of 30 (Start time: 12:41:46 PM, 30 June 2021)

According to determinant properties, the determinants of resulting matrix equal to k delta if elements of rows are

Select the correct option

- Divided to constant k
- Multiplied to constant k
- Added to constant k
- Subtracting to constant k

Click to Save Answ

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Quiz Start Time: 12:27 PM, 30 Jun

Question # 27 of 30 (Start time: 12:42:21 PM, 30 June 2021)

Total M

For the Linear Equation : $x_1 + x_2 + x_3 + x_4 = 1$, how many Parametres t_i's would have to introduce in order to find its solution?

Select the correct option

Reload Math Equa

- 2
- 4
- 1
- 3 ●

Click to Save Answer & Move to Next Ques

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Quiz Start Time: 1

Question # 28 of 30 (Start time: 12:42:48 PM, 30 June 2021)

For the following system of Equations:

$x + 2y = 0, 2x - y = 0;$

which variable can be taken as free?

Select the correct option

Rel

- only y
- both x and y
- only x
- either x or y ●

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Quiz Start Tir

Question # 29 of 30 (Start time: 12:43:41 PM, 30 June 2021)

If a multiple of one row of a square matrix A is added to another row to produce a matrix B, then which of the following condition is true?

Select the correct option

- detB = detA
- detB = k detA
- detA detB = 0
- detA detB = detA

Click to Save Answer &

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Question # 30 of 30 (Start time: 12:44:01 PM, 30 June 2021)

In Jacobi's method, the diagonal entries of a matrix are _____.

Select the correct option

- zero
- all nonzero
- all zero
- any constant

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Question # 1 of 30 (Start time: 01:00:37 PM, 30 June 2021)

Algebra is a branch of Mathematics that uses -------- statements to describe relationship between things.

Select the correct option

- Mathematical
- Theoretical

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Question # 2 of 30 (Start time: 01:01:00 PM, 30 June 2021)

$$\begin{pmatrix} Cos\theta & Sin\theta \\ -Sin\theta & Cos\theta \end{pmatrix}^{-1} =$$

Select the correct option

- $\begin{pmatrix} Cos\theta & Sin\theta \\ -Sin\theta & Cos\theta \end{pmatrix}$
- $\begin{pmatrix} Cos\theta & -Sin\theta \\ Sin\theta & Cos\theta \end{pmatrix}$
- $\begin{pmatrix} -Cos\theta & Sin\theta \\ Sin\theta & Cos\theta \end{pmatrix}$
- $\begin{pmatrix} Cos\theta & Sin\theta \\ Sin\theta & -Cos\theta \end{pmatrix}$

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Quiz Sta

Question # 3 of 30 (Start time: 01:01:29 PM, 30 June 2021)

Which of the following would be the value of t, if $\begin{bmatrix} 2e^{-3t} \\ 5e^{-3t} \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \end{bmatrix}$?

Select the correct option

- zero
- $\frac{\begin{bmatrix} 2 \\ 5 \end{bmatrix}}{\begin{bmatrix} 2 \\ 5 \end{bmatrix}} = 1$
- $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$
- $\pm\frac{1}{3}$

Click to Save Answ

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Question # 4 of 30 (Start time: 01:01:54 PM, 30 June 2021)

If B is the matrix obtained by interchanging two rows or columns of the square matrix A, then det (B) = _____.

Select the correct option

- det(A)
- 1/det(A)
- - det(A)
- -1/det(A)

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Question # 5 of 30 (Start time: 01:02:21 PM, 30 June 2021)

If n>m, then the set $\{a_1, a_2, \ldots, a_n\}$ of vectors in R^m must be ---------

Select the correct option

- ○ Linearly Independent
- ● Linearly Dependent

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Question # 6 of 30 (Start time: 01:02:42 PM, 30 June 2021)

Let V be a five dimensional vector space, and let S be a subset of V which spans V, then S

Select the correct option

- ○ Must be linearly dependent,
- ○ Must be basis for V
- ● Must consist of at least five elements
- ○ Must have exactly five elements

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Question # 7 of 30 (Start time: 01:03:04 PM, 30 June 2021)

The determinant of the identity matrix is _____ .

Select the correct option

- 0
- -1
- n x (number of rows or coulms)
- 1

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MTH501:Grand Quiz Quiz Start Time: 01:00 PM, 30 June 20

Question # 8 of 30 (Start time: 01:03:40 PM, 30 June 2021) Total Marks

If a system of equations is solved using the Jacobi's method , then which of the following is the most appropriate answer about the matrix M that is derived from the coefficient matrix ?

Select the correct option

- All of its entries on the diagonal must be zero.
- All of its entries below the diagonal must be zero.
- All of its entries below and above the diagonal must be zero.
- All of its entries above the diagonal must be zero.

Click to Save Answer & Move to Next Question

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Quiz Start Time: 01:00 PM, 30 June 202

Question # 9 of 30 (Start time: 01:04:08 PM, 30 June 2021)

Total Marks:

Let P_n be the set of real valued function of the form $P(x) = a_0 + a_1 x + a_2 x^2 + \ldots + a_n x^n$ wheren is a non-negative integer and $a_0, a_1, \ldots, a_n$ are real numbers, then P_n is a ---------- $F(-\infty, + + \infty)$.

Select the correct option

Reload Math Equations

- (●) Subspace
- ( ) Space
- ( ) Vector Space
- ( ) None of the above

Click to Save Answer & Move to Next Question

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Quiz S

Question # 10 of 30 (Start time: 01:04:33 PM, 30 June 2021)

The system of equations :

$x_1 = 1,$

$x_2 = 0;$ can be expressed in the form - - - - - .

Select the correct option

- (●) $Ax = b$
- ( ) $Ax = 0$
 $By = 1$
- ( ) $Ax = 1$
 $By = 0$
- ( ) $Ax = 0$

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Quiz Start Ti

Question # 11 of 30 (Start time: 01:04:57 PM, 30 June 2021)

For any subspace W of a vector space V. which one is not the axiom for subspace.

Select the correct option

- For all u, v in W and u.v must be in W.
- 0 must be in W.
- For all u, v in W and u - v must be in W.
- For any scalar k and u in W then k.u in W.

Click to Save Answer &

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Quiz Sta

Question # 12 of 30 (Start time: 01:05:22 PM, 30 June 2021)

If M is a square matrix having two rows equal then which of the following about the determinant of the matrix is true?

Select the correct option

- ● det (M)=0
- ○ det (M) is not equal to '1'
- ○ det (M)=1
- ○ det (M) is not equal to '0'

Click to Save Answ

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Quiz Start Time: 01:00 PM

Question # 13 of 30 (Start time: 01:06:04 PM, 30 June 2021)

Which of the following Elementary Row operations would perform in order to get $\begin{pmatrix} 1 & 2 & -5 \\ -4 & 1 & -6 \\ 6 & 3 & -4 \end{pmatrix} \sim \begin{pmatrix} 1 & 2 & -5 \\ 0 & 9 & -26 \\ 0 & -9 & 26 \end{pmatrix}$?

Select the correct option

Reload M

- $R_2' \to R_2 - 4R_1, R_3' \to R_3 + 6R_1$
- $R_2' \to R_1 + 4R_2, R_3' \to R_1 - 6R_3$
- (●) $R_2' \to R_2 + 4R_1, R_3' \to R_3 - 6R_1$
- $R_2' \to R_1 - 4R_2, R_3' \to R_1 + 6R_3$

Click to Save Answer & Move to N

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Question # 14 of 30 (Start time: 01:06:43 PM, 30 June 2021)

A matrix A and its transpose have the same determinant.

Select the correct option

- (●) TRUE
- ( ) FALSE

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Question # 15 of 30 (Start time: 01:07:07 PM, 30 June 2021)

A partitioned matrix is a partition of a matrix into rectangular smaller matrices called _____.

Select the correct option

- element
- blocks
- row
- column

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Question # 16 of 30 (Start time: 01:07:36 PM, 30 June 2021)

The subspace {0} in a vector space V is known as the:

Select the correct option

- ○ Subspace
- ○ Basis
- ○ Spanning set
- ● Zero vector space

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Question # 17 of 30 (Start time: 01:08:01 PM, 30 June 2021)

If A is an invertible square matrix then

Select the correct option

- $(A^T)^{-1} = (A^{-1})^T$
- $(A^T)^{-1} = (A^{-1})^{-1}$
- $(A^T)^T = (A^{-1})^T$
- None of the above.

Click to Save

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Quiz Start Time:

Question # 18 of 30 (Start time: 01:08:22 PM, 30 June 2021)

Set $\left\{\begin{pmatrix}1\\2\end{pmatrix}, \begin{pmatrix}0\\0\end{pmatrix}\right\}$ is Linearly - - - - - in $\mathbb{R}^2$.

Select the correct option

- Dependent
- Independent

Click to Save Answer & Mo

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Quiz Start Time: 01:00 PM, 30 June 2021

Question # 19 of 30 (Start time: 01:08:47 PM, 30 June 2021)

Total Marks: 1

Under which of the following condition, a system of Linear Equations whose Row - Reduce form is $\left(\begin{array}{cc|c}1 & 2 & -1\\ 0 & 0 & h-3k\end{array}\right)$ has $Infinite$ many solutions?

Select the correct option

Reload Math Equations

- $h \neq 3k$
- $(h,k) \neq (0,0)$
- $(h,k) = (0,0)$
- $h = 3k$

Click to Save Answer & Move to Next Question

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Quiz Start Time: 01:00 PM

Question # 20 of 30 (Start time: 01:09:32 PM, 30 June 2021)

Which of the following is corresponding Matrix form of the Linear equation $x + y = 3$?

Select the correct option

Reload M

- [x] $x\,[1] + y\,[1] = 3\,[1]$
- [ ] $x\begin{pmatrix}1\\0\end{pmatrix} + y\begin{pmatrix}0\\1\end{pmatrix} = 3\begin{pmatrix}0\\1\end{pmatrix}$
- [ ] $x\begin{pmatrix}1\\0\end{pmatrix} + y\begin{pmatrix}0\\1\end{pmatrix} = 3\begin{pmatrix}1\\0\end{pmatrix}$
- [ ] Cant be expressed in Matrix form.

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Quiz Start Time: 01:0

Question # 21 of 30 (Start time: 01:09:56 PM, 30 June 2021)

$$[8 \quad -4 \quad 5] \times \begin{bmatrix}3\\2\\-1\end{bmatrix} =$$

Select the correct option

Reloa

- [ ] $[24 \quad -8 \quad -5]$
- [ ] $\begin{pmatrix}24\\-8\\-5\end{pmatrix}$
- [x] 11
- [ ] $\begin{pmatrix}24 & -12 & 15\\16 & -8 & 10\\-8 & 4 & -5\end{pmatrix}$

Click to Save Answer & Move t

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Quiz Start Time: 01:

Question # 22 of 30 (Start time: 01:10:34 PM, 30 June 2021)

The equation : $0x - 0y = 0$ has - - - - solution(s).

Select the correct option

Relo

- No
- Infinite many
- Unique
- multiple finite

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Quiz Start T

Question # 22 of 30 (Start time: 01:10:34 PM, 30 June 2021)

The equation : $0x - 0y = 0$ has - - - - solution(s).

Select the correct option

- No
- Infinite many
- Unique
- multiple finite

Click to Save Answer &

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Question # 23 of 30 (Start time: 01:11:29 PM, 30 June 2021)

Linear algebra is the study of linear sets of equations and their ---------- properties.

Select the correct option

- transformation
- non-linear

Click to Save A

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Question # 24 of 30 (Start time: 01:12:12 PM, 30 June 2021)

Gauss-Seidel method is similar to the

Select the correct option

- (●) Jacobi Method
- ( ) Iteration Method
- ( ) Regula-Falsi Method
- ( ) Newton Raphson Method

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Question # 25 of 30 (Start time: 01:12:39 PM, 30 June 2021)

Total Marks:

Under which of the following condition, a system of Linear Equations whose Row - Reduce form is $\left(\begin{array}{cc|c} 1 & 2 & -1 \\ 0 & 0 & h-3k \end{array}\right)$ has NO solution?

Select the correct option

Reload Math Equations

- ( ) $(h,k) \neq (0,0)$
- (●) $h \neq 3k$
- ( ) $(h,k) = (0,0)$
- ( ) $h = 3k$

Click to Save Answer & Move to Next Question

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Question # 26 of 30 (Start time: 01:13:08 PM, 30 June 2021)

The two vectors $(-2, 1)$ and $(1, 2)$ are

Select the correct option

○	Pointing in the opposite direction of each other
○	Linearly dependent of each other
●	Perpendicular to each other
○	None of the above

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Question # 27 of 30 (Start time: 01:13:58 PM, 30 June 2021)

If the Gauss-Seidel method is convergent , then the coefficient matrix is _____.

Select the correct option

- strictly diagonally dominant
- not strictly diagonally dominant
- not invertible
- singular

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Quiz Start Tir

Question # 28 of 30 (Start time: 01:14:22 PM, 30 June 2021)

If A is invertible and b in R^n be any vector. Then, we must have a matrix $A^{-1}b$,which is a solution of _____.

Select the correct option

- $A^t x = b$
- $Ax = b$
- $A^{-1}x = b$
- $A^2 x = b$

Click to Save Answer &

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$$x_1 + 2x_2 + 3x_3 = 7$$
$$4x_1 + x_2 + 2x_3 = 2$$
$$-4x_1 + 3x_2 + 9x_3 = 4$$

The augmented matrix for the system is

Select the correct option

Reload Ma

- $\begin{bmatrix} 1 & 2 & 3 & 7 \\ -4 & 1 & 2 & 2 \\ 4 & 3 & 9 & 4 \end{bmatrix}$
- $\begin{pmatrix} 1 & 2 & 3 \\ 4 & 1 & 2 \\ -4 & 3 & 9 \end{pmatrix}$
- $\begin{bmatrix} 1 & 2 & 3 \\ -4 & 1 & 2 \\ 4 & 3 & 9 \end{bmatrix}$
- $\begin{bmatrix} 1 & 2 & 3 & 7 \\ 4 & 1 & 2 & 2 \\ -4 & 3 & 9 & 4 \end{bmatrix}$

Click to Save Answer & Move to Nex

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MTH501:Grand Quiz

Quiz Start Time:

Question # 29 of 30 (Start time: 01:14:56 PM, 30 June 2021)

$$x_1 + 2x_2 + 3x_3 = 7$$
$$4x_1 + x_2 + 2x_3 = 2$$
$$-4x_1 + 3x_2 + 9x_3 = 4$$

The augmented matrix for the system is

Select the correct option

- $\begin{bmatrix} 1 & 2 & 3 & 7 \\ -4 & 1 & 2 & 2 \\ 4 & 3 & 9 & 4 \end{bmatrix}$
- $\begin{pmatrix} 1 & 2 & 3 \\ 4 & 1 & 2 \\ -4 & 3 & 9 \end{pmatrix}$
- $\begin{bmatrix} 1 & 2 & 3 \\ -4 & 1 & 2 \\ 4 & 3 & 9 \end{bmatrix}$
- $\begin{bmatrix} 1 & 2 & 3 & 7 \\ 4 & 1 & 2 & 2 \end{bmatrix}$

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MTH501:Grand Quiz

Quiz Start Time

Question # 30 of 30 (Start time: 01:15:51 PM, 30 June 2021)

Which of the following is true for the speed of convergence of both the Jacobi and Gauss-Seidel sequences?

Select the correct option

- It depends on how much the diagonal entries dominate the preceding row sums.
- It depends on how much the diagonal entries dominate the preceding column sums.
- It depends on how much the diagonal entries dominate the corresponding row sums.
- It depends on how much the diagonal entries dominate the corresponding column sums.

Click to Save Answer & M

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Quiz Start Time

Question # 30 of 30 (Start time: 01:15:51 PM, 30 June 2021)

Which of the following is true for the speed of convergence of both the Jacobi and Gauss-Seidel sequences?

Select the correct option

- It depends on how much the diagonal entries dominate the preceding row sums.
- It depends on how much the diagonal entries dominate the preceding column sums.
- It depends on how much the diagonal entries dominate the corresponding row sums.
- It depends on how much the diagonal entries dominate the corresponding column sums.

Click to Save Answer & M

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MTH501:Grand Quiz

Question # 1 of 30 (Start time: 01:24:29 PM, 30 June 2021)

What is the maximum possible number of pivots in a 6x6 matrix ?

Select the correct option

- 4
- 0
- 2
- 6

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MTH501:Grand Quiz

Question # 2 of 30 (Start time: 01:25:05 PM, 30 June 2021)

The subspace {0} in a vector space V is known as the:

Select the correct option

- Spanning set
- Subspace
- Basis
- Zero vector space

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MTH501:Grand Quiz

Question # 3 of 30 (Start time: 01:25:36 PM, 30 June 2021)

Which of the following is true for the speed of convergence of both the Jacobi and Gauss-Seidel sequences?

Select the correct option

- It depends on how much the diagonal entries dominate the corresponding row sums.
- It depends on how much the diagonal entries dominate the preceding row sums.
- It depends on how much the diagonal entries dominate the corresponding column sums.
- It depends on how much the diagonal entries dominate the preceding column sums.

Click to Sav

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MTH501:Grand Quiz

Question # 4 of 30 (Start time: 01:26:01 PM, 30 June 2021)

The word algorithm comes from the famous Muslim mathematician

Select the correct option

- Al-Kindi
- Al-Khwarizimi
- None of these
- Omar Khayyam

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MTH501:Grand Quiz

Quiz Sta

Question # 5 of 30 (Start time: 01:26:24 PM, 30 June 2021)

Any set

$$\{v_1, v_2, \ldots, v_p\} \in R^n$$

is linearly dependent if

Select the correct option

- p>n
- p<n

Click to Save Answ

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MTH501:Grand Quiz

Question # 6 of 30 (Start time: 01:26:44 PM, 30 June 2021)

Cramer's rule is a formula for solving systems of equations by _____.

Select the correct option

- matrix inversion
- comparison
- substitutions
- determinants

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MTH501 - Linear Algebra

MTH501:Grand Quiz

Quiz Start Time:

Question # 7 of 30 (Start time: 01:27:07 PM, 30 June 2021)

Which of the following would be the value of t, if $\begin{bmatrix} 2e^{-3t} \\ 5e^{-3t} \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \end{bmatrix}$?

Select the correct option

R

- $\frac{\begin{bmatrix} 2 \\ 5 \end{bmatrix}}{\begin{bmatrix} 2 \\ 5 \end{bmatrix}} = 1$
- $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$
- zero
- $\pm\frac{1}{3}$

Click to Save Answer & Mo

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MTH501:Grand Quiz

Question # 8 of 30 (Start time: 01:27:33 PM, 30 June 2021)

A null space is a vector space.

Select the correct option

- ● True
- ○ False

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MTH501:Grand Quiz

Quiz Start Time: 01:24 PM

Question # 9 of 30 (Start time: 01:28:01 PM, 30 June 2021)

Which of the following is corresponding Matrix form of the Linear equation $x + y = 3$?

Select the correct option

Reload Ma

- ○ $x\begin{pmatrix}1\\0\end{pmatrix} + y\begin{pmatrix}0\\1\end{pmatrix} = 3\begin{pmatrix}0\\1\end{pmatrix}$
- ○ Cant be expressed in Matrix form.
- ● $x\,[1] + y\,[1] = 3\,[1]$
- ○ $x\begin{pmatrix}1\\0\end{pmatrix} + y\begin{pmatrix}0\\1\end{pmatrix} = 3\begin{pmatrix}1\\0\end{pmatrix}$

Click to Save Answer & Move to Ne

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Quiz

Question # 10 of 30 (Start time: 01:28:31 PM, 30 June 2021)

If a matrix equation is solved using the Jacobi's method, then which of the following is true about its main diagonal?

Select the correct option

- It has two zeros along its main diagonal.
- It has more than two zeros along its main diagonal.
- It has no zeros along its main diagonal.
- It has at least one zero along its main diagonal.

Click to Save A

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MTH501 - Linear Algebra

MTH501:Grand Quiz

Quiz Start Time: 01:24 PM

Question # 11 of 30 (Start time: 01:28:58 PM, 30 June 2021)

Let A and B be the square matrices. Then, A and B are invertible with $B = A^{-1}$ and $A = B^{-1}$ if and only if $AB = BA$ equals to a (an) _____ matrix.

Select the correct option

Reload Ma

- identity
- rectangular
- square
- singular

Click to Save Answer & Move to Ne

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MTH501:Grand Quiz

Question # 12 of 30 (Start time: 01:29:19 PM, 30 June 2021)

A blocked matrix in which blocks (blocked matrices) are repeated down the diagonals of the matrix is called a _____ matrix.

Select the correct option

- blocked rectangular
- blocked square
- blocked diagonal-constant
- blocked identity

Click to Sav

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MTH501 - Linear Algebra

MTH501:Grand Quiz

Quiz Start Time: 0

Question # 13 of 30 (Start time: 01:29:43 PM, 30 June 2021)

As soon as an approximate solution of a linear system is sufficiently accurate for practical work then which of the following is true for an iterative process?

Select the correct option

- (●) The process may be stopped.
- ( ) The process may be carried on.
- ( ) The row-echelon method may be used.
- ( ) The reduced row-echelon method may be used.

Click to Save Answer & Move

MTH501:Grand Quiz

Quiz Start Ti

Question # 14 of 30 (Start time: 01:30:07 PM, 30 June 2021)

Which of the following will be the Linear Combinition corresponding to $\begin{pmatrix} -2 & 3 \\ 5 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$?

Select the correct option

- ( ) $\begin{pmatrix} 3 \\ 1 \end{pmatrix} x + \begin{pmatrix} -2 \\ 5 \end{pmatrix} y$
- (●) $\begin{pmatrix} -2 \\ 5 \end{pmatrix} x + \begin{pmatrix} 3 \\ 1 \end{pmatrix} y$
- ( ) $\begin{pmatrix} -2 \\ 3 \end{pmatrix} x + \begin{pmatrix} 5 \\ 1 \end{pmatrix} y$
- ( ) $\begin{pmatrix} 5 \\ 1 \end{pmatrix} x + \begin{pmatrix} -2 \\ 3 \end{pmatrix} y$

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MTH501:Grand Quiz

Question # 15 of 30 (Start time: 01:30:31 PM, 30 June 2021)

If A is strictly diagonally dominant matrix, then which of the following is true for A?

Select the correct option

- It is a null matrix.
- It is an invertible matrix.
- It is a rectangular matrix.
- It is a singular matrix.

Click to Save

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MTH501:Grand Quiz

Question # 16 of 30 (Start time: 01:31:00 PM, 30 June 2021)

All the lines those passes through origin are not the subspace of a plane.

Select the correct option

- TRUE
- FALSE

Clic

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MTH501:Grand Quiz

Question # 17 of 30 (Start time: 01:31:26 PM, 30 June 2021)

The transpose of the sum of two matrices is equal to

Select the correct option

- ● $(A+B)^T = A^T + B^T$
- ○ $(A+B)^T = B^T A^T$

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MTH501:Grand Quiz

Quiz Start Time:

Question # 18 of 30 (Start time: 01:31:45 PM, 30 June 2021)

The values of x and y which satisfy the matrix equation : $\begin{pmatrix} 4 \\ y \end{pmatrix} = x \begin{pmatrix} 2 \\ 3 \end{pmatrix}$ *are* $----$.

Select the correct option

- ○ $\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 6 \\ 2 \end{pmatrix}$
- ○ $\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$
- ● $\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2 \\ 6 \end{pmatrix}$
- ○ $\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$

Click to Save Answer & M

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Question # 19 of 30 (Start time: 01:32:11 PM, 30 June 2021)

In matrix factorization A=LU, where L is a lower triangular matrix with any _____ on the diagonal and U is an upper triangular matrix.

Select the correct option

- fixed variable
- fixed constant
- constant
- variable

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MTH501:Grand Quiz

Question # 20 of 30 (Start time: 01:32:41 PM, 30 June 2021)

The transpose of the product of two matrices is equal to

Select the correct option

- (●) $(AB)^T = B^T A^T$
- ( ) $(AB)^T = A^T B^T$

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MTH501:Grand Quiz

Quiz Start Time: 01:2

Question # 21 of 30 (Start time: 01:33:06 PM, 30 June 2021)

$\begin{pmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}$ is an example of - - - - - - - matrix.

Select the correct option

Reloa

- ( ) Column
- (●) Zero
- ( ) Non-Singular
- ( ) Singular

Click to Save Answer & Move t

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MTH501:Grand Quiz

Question # 22 of 30 (Start time: 01:33:32 PM, 30 June 2021)

What is the maximum possible number of pivots in a '3 by 3' matrix ?

Select the correct option

- 3
- 0
- 5
- 1

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MTH501 - Linear Algebra

MTH501:Grand Quiz

Quiz Start Time: 01:2

Question # 23 of 30 (Start time: 01:33:59 PM, 30 June 2021)

Set $\left\{\begin{pmatrix}1\\2\end{pmatrix}, \begin{pmatrix}0\\0\end{pmatrix}\right\}$ is Linearly - - - - - in $\mathbb{R}^2$.

Select the correct option

Reloa

- Dependent
- Independent

Click to Save Answer & Move t

MTH501:Grand Quiz

Quiz Start Time: 01:24 I

Question # 24 of 30 (Start time: 01:34:32 PM, 30 June 2021)

If $\begin{vmatrix} \frac{d}{dx}e^{x} & \frac{d}{dx}e^{-x} \\ \frac{d}{dx}e^{-x} & \frac{d}{dx}e^{x} \end{vmatrix} = 0,$ then $x = - - -$

Select the correct option

Reload

- -e
- e
- 1
- 0

Click to Save Answer & Move to

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MTH501 - Linear Algebra



MTH501:Grand Quiz

Quiz St

Question # 25 of 30 (Start time: 01:34:56 PM, 30 June 2021)

According to determinant properties, the determinants of resulting matrix equal to k delta if elements of rows are

Select the correct option

- ○ Subtracting to constant k
- ○ Divided to constant k
- ● Multiplied to constant k
- ○ Added to constant k

Click to Save Answ

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MTH501:Grand Quiz

Question # 26 of 30 (Start time: 01:35:27 PM, 30 June 2021)

In matrices, rows are denoted by -------

Select the correct option

- C
- R
- A
- B

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MTH501:Grand Quiz

Quiz Start Time: 01:24 PM, 30 June 2021

Question # 27 of 30 (Start time: 01:35:50 PM, 30 June 2021)

Total Marks: 1

A transformation $T : R^n \rightarrow R^m$ Is a rule that assigns to each vector x in R^n , an image vector T(x) in R^m. The set R^n is called the_____ , and R^m is called the _____.

Select the correct option

Reload Math Equations

- domain of T, co-domain of T
- co-domain of T, domain of T

Click to Save Answer & Move to Next Question

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MTH501:Grand Quiz

Question # 28 of 30 (Start time: 01:36:16 PM, 30 June 2021)

For a non-singular matrix A, A^{-1} is equal to

Select the correct option

- ● $Adj(A)/det(A)$
- ○ None of the above.
- ○ $det(A) * Adj(A)$
- ○ $det(A) * A$

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MTH501:Grand Quiz

Question # 29 of 30 (Start time: 01:36:39 PM, 30 June 2021)

Jacobi's method assumes that the diagonal entries of an invertible matrix A are all

Select the correct option

- zero
- Non-zero

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MTH501:Grand Quiz

Quiz Start Time: 0

Question # 30 of 30 (Start time: 01:37:02 PM, 30 June 2021)

If r and s are vectors in V and a = (r, s) and b = (u,v) then which of the following is true for addition in V?

Select the correct option

- r+s=(r+s, u+v)
- r+s=(r,u, s,v)
- r+s=(r+u , s+v) (selected)
- r+s=(u,v,s,r)

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MTH501:Grand Quiz

Question # 1 of 30 (Start time: 01:57:37 PM, 30 June 2021)

If A is strictly diagonally dominant matrix, then which of the following is true for A?

Select the correct option

- It is a singular matrix.
- It is a rectangular matrix.
- It is a null matrix.
- It is an invertible matrix.

Click to

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MTH501:Grand Quiz

Question # 2 of 30 (Start time: 01:58:18 PM, 30 June 2021)

Inverse of a matrix is given by

Select the correct option

- $A^{-1} = \frac{1}{detA} adj(A)$
- $A^{-1} = \frac{1}{detA} A^{-1}$

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MTH501:Grand Quiz

Question # 3 of 30 (Start time: 01:58:39 PM, 30 June 2021)

We can subtract two matrices A and B if their--------

Select the correct option

- ○ Rows are same
- ○ Elements are same
- ● Order is same
- ○ Columns are same

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MTH501:Grand Quiz

Question # 4 of 30 (Start time: 01:59:03 PM, 30 June 2021)

The transpose of the product of two matrices is equal to

Select the correct option

- ● $(AB)^T = B^T A^T$
- ○ $(AB)^T = A^T B^T$

MTH501:Grand Quiz Quiz Start Tim

Question # 5 of 30 (Start time: 01:59:21 PM, 30 June 2021)

$$\begin{pmatrix} Cos\theta & Sin\theta \\ -Sin\theta & Cos\theta \end{pmatrix}^{-1} =$$

Select the correct option

- ● $\begin{pmatrix} Cos\theta & -Sin\theta \\ Sin\theta & Cos\theta \end{pmatrix}$
- ○ $\begin{pmatrix} -Cos\theta & Sin\theta \\ Sin\theta & Cos\theta \end{pmatrix}$
- ○ $\begin{pmatrix} Cos\theta & Sin\theta \\ -Sin\theta & Cos\theta \end{pmatrix}$
- ○ $\begin{pmatrix} Cos\theta & Sin\theta \\ Sin\theta & -Cos\theta \end{pmatrix}$

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Question # 6 of 30 (Start time: 01:59:44 PM, 30 June 2021)

Any system of n linear equations in (n+1) variable can be solved by Cramer's rule.

Select the correct option

- TRUE
- FALSE

Click t

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MTH501:Grand Quiz

Question # 7 of 30 (Start time: 02:00:06 PM, 30 June 2021)

What is the maximum possible number of pivots in a 6x6 matrix ?

Select the correct option

- 0
- 4
- 2
- 6

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MTH501:Grand Quiz

Question # 8 of 30 (Start time: 02:00:29 PM, 30 June 2021)

The equation $Ax = b$ has a solution if and only if

Select the correct option

- b is a non-linear combination of the columns of A
- b is the linear combination of the columns of A

Click to S

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MTH501:Grand Quiz

Quiz Start Time: 01:57

Question # 9 of 30 (Start time: 02:00:52 PM, 30 June 2021)

In LU-decomposition, if A is the square matrix of order m, then the order of both lower and upper triangular matrices will be _____.

Select the correct option

Reload

- $m + 1$
- m
- m^2
- $2m$

Click to Save Answer & Move to

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MTH501:Grand Quiz

Question # 10 of 30 (Start time: 02:02:07 PM, 30 June 2021)

Which of the following is basic unit of an Algebraic expression?

Select the correct option

- Term
- Number
- Constant
- Variable

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MTH501:Grand Quiz

Quiz Sta

Question # 11 of 30 (Start time: 02:02:29 PM, 30 June 2021)

If $i = \sqrt{-1} \in \mathbb{C}$, and $n \in \mathbb{Z}^+$, then $\begin{pmatrix} i & 0 \\ 0 & i \end{pmatrix}^{4n} =$ – – – – – ––.

Select the correct option

- $\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$
- $\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$
- $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$
- $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

Click to Save Answ

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MTH501:Grand Quiz

Question # 12 of 30 (Start time: 02:03:14 PM, 30 June 2021)

We can add the matrices of _____.

Select the correct option

- same number of columns
- same order
- same number of rows.
- different order

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MTH501:Grand Quiz

Question # 13 of 30 (Start time: 02:03:40 PM, 30 June 2021)

If all the entries of a row or a column of a square matrix are zero, then det (A) will be _____.

Select the correct option

- one
- non zero
- infinity
- zero

Click

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MTH501:Grand Quiz

Quiz Start Ti

Question # 14 of 30 (Start time: 02:04:05 PM, 30 June 2021)

The solution set of system of equations as AX=B is calculated only when matrix equations both sides are Subtracted by A^{-1}

Select the correct option

- Divided A^{-1}
- Subtracted A^{-1}
- Multiplied by A^{-1}
- Added by A^{-1}

Click to Save Answer &

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Question # 15 of 30 (Start time: 02:04:38 PM, 30 June 2021)

$$\begin{vmatrix} \int_0^1 e^x dx & \int_0^1 e^{1-x} dx \\ -\int_0^1 e^{1-x} dx & \int_0^1 e^x dx \end{vmatrix} = - - - -$$

Select the correct option

- $(e-1)^2$
- 0
- $2(e-1)^2$
- $e-1$

Click to Sa

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Question # 16 of 30 (Start time: 02:05:03 PM, 30 June 2021)

Let A be a square nxn matrix, then which of the following statement(s) is(are) equivalent?

Select the correct option

- (b) A^T is an invertible matrix.
- (c) The equation Ax = 0 has only the trivial solution.
- (d) Options (a to c).
- (a) A is an invertible matrix.

Click to S

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MTH501:Grand Quiz Qui

Question # 17 of 30 (Start time: 02:05:37 PM, 30 June 2021)

Vector $\begin{pmatrix} -1 \\ 0 \end{pmatrix}$ is Linearly $------$.

Select the correct option

- Dependent
- Independent

Click to Save A

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MTH501:Grand Quiz

Quiz Sta

Question # 18 of 30 (Start time: 02:06:04 PM, 30 June 2021)

A set of two vectors

$$\{v_1, v_2\}$$

is linearly dependent if and only if one of the vector is

Select the correct option

- Additive inverse of the other
- Multiple of the other

Click to Save Answ

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MTH501:Grand Quiz

Question # 19 of 30 (Start time: 02:06:29 PM, 30 June 2021)

The word algorithm comes from the famous Muslim mathematician

Select the correct option

- Al-Kindi
- Al-Khwarizimi
- None of these
- Omar Khayyam

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MTH501:Grand Quiz

Question # 20 of 30 (Start time: 02:06:57 PM, 30 June 2021)

If B is the matrix obtained by interchanging two rows or columns of the square matrix A, then det (B) = _____.

Select the correct option

- ● - det(A)
- ○ -1/det(A)
- ○ 1/det(A)
- ○ det(A)

Click to S

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MTH501:Grand Quiz

Quiz Start Time: 0

Question # 21 of 30 (Start time: 02:07:19 PM, 30 June 2021)

Which of the following system will have the non - trivial solution?

Select the correct option

Re

- [x] $x = y = 1$
- [ ] $x = y = 0$

Click to Save Answer & Mov

MTH501:Grand Quiz

Question # 22 of 30 (Start time: 02:07:40 PM, 30 June 2021)

Gauss-Seidel method is also termed as a method of

Select the correct option

- [ ] Iteration Method
- [ ] Elimination Method
- [x] Successive Displacement
- [ ] False Position Method

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MTH501 - Linear Algebra

MTH501:Grand Quiz

Quiz Start Time: 01:57 PM, 30 June 2021

Question # 23 of 30 (Start time: 02:08:04 PM, 30 June 2021) **Total Marks:** 1

If a system of equations is solved using the Jacobi's method , then which of the following is the most appropriate answer about the matrix M that is derived from the coefficient matrix ?

Select the correct option

- All of its entries on the diagonal must be zero.
- All of its entries below and above the diagonal must be zero.
- All of its entries below the diagonal must be zero.
- All of its entries above the diagonal must be zero.

Click to Save Answer & Move to Next Question

MTH501:Grand Quiz

Quiz Start Time: 01:57 P

Question # 24 of 30 (Start time: 02:08:31 PM, 30 June 2021)

A partitioned square matrix 'A' is said to be _____ matrix if the matrices on the main diagonal are square and all matrices below the main diagonal are zero.

Select the correct option

- block lower triangular
- diagonal-constant
- block upper triangular
- null

Click to Save Answer & Move to N

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MTH501:Grand Quiz

Quiz Start Time

Question # 25 of 30 (Start time: 02:08:54 PM, 30 June 2021)

A homogeneous linear system always has the trivial solution; there are only two possibilities for its solutions:

Select the correct option

- ○ The system has infinitely many solutions in addition to the trivial solution.
- ● Both (a) and (b)
- ○ The system has only the trivial solution.
- ○ None of the above

Click to Save Answer & M

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MTH501:Grand Quiz

Quiz Start

Question # 26 of 30 (Start time: 02:09:17 PM, 30 June 2021)

An m x m _____ matrix is only square matrix that is both unit lower triangular and upper triangular.

Select the correct option

- diagonal
- scalar
- null
- identity

MTH501:Grand Quiz

Quiz Start Time: 01:57

Question # 27 of 30 (Start time: 02:09:43 PM, 30 June 2021)

If A is a square matrix, then the Minor of entry ith row and jth column is to be the determinant of the sub matrix that remains when the ith row and jth column of A are:

Select the correct option

- multiplied
- added
- divided
- deleted

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MTH501:Grand Quiz

Question # 28 of 30 (Start time: 02:10:07 PM, 30 June 2021)

What is 'a' if

$$B = \begin{pmatrix} 1 & 4 \\ 2 & a \end{pmatrix}$$

is a singular matrix ?

Select the correct option

- 7
- 8
- 6
- 5

Click to S

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MTH501:Grand Quiz

Question # 29 of 30 (Start time: 02:10:32 PM, 30 June 2021)

The columns of a matrix A are ---------- if and only if the equation $Ax = 0$ has only the trivial solution.

Select the correct option

- linearly dependent
- linearly independent

Click

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MTH501:Grand Quiz

Quiz St

Question # 30 of 30 (Start time: 02:10:56 PM, 30 June 2021)

Vectors $\begin{pmatrix} -2 \\ 3 \end{pmatrix}$ and $\begin{pmatrix} -66 \\ 99 \end{pmatrix}$ are linearly $------$.

Select the correct option

- Independent
- Dependent

Click to Save Answ

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MTH501:Grand Quiz

Question # 1 of 30 (Start time: 02:12:51 PM, 30 June 2021)

According to determinant properties, the determinant equals to zero if row is

Select the correct option

- Divided by row
- Divided by column
- Multiplied to row
- Multiplied to column

Clic

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MTH501:Grand Quiz

Quiz

Question # 2 of 30 (Start time: 02:13:22 PM, 30 June 2021)

$(AB)^t = ?$

Select the correct option

- BA
- $B^t A^t$

Click to Save A

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MTH501:Grand Quiz

Quiz Sta

Question # 3 of 30 (Start time: 02:13:46 PM, 30 June 2021)

By using determinants, we can easily check that the solution of the given system of linear equation exits and it is unique.

Select the correct option

- ● TRUE
- ○ FALSE

Click to Save Answ

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Quiz Start Tim

Question # 4 of 30 (Start time: 02:14:05 PM, 30 June 2021)

A partitioned matrix 'A' is said to be block diagonal if the matrices on the main diagonal are square and all other position matrices are _____.

Select the correct option

- ○ unit
- ● zero
- ○ nonzero skew symmetric
- ○ nonzero symmetric

Click to Save Answer & M

MTH501:Grand Quiz

Quiz Start Time: 02:12 PM, 30 Jun

Question # 5 of 30 (Start time: 02:14:30 PM, 30 June 2021)

Total M

For the Linear Equation : $x_1 + x_2 + x_3 + x_4 = 1$, how many Parametres t_i's would have to introduce in order to find its solution?

Select the correct option

Reload Math Equa

- ○ 1
- ● 3
- ○ 4
- ○ 2

Click to Save Answer & Move to Next Ques

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MTH501:Grand Quiz

Question # 6 of 30 (Start time: 02:14:59 PM, 30 June 2021)

For any subspace W of a vector space V. which one is not the axiom for subspace.

Select the correct option

- For all u, v in W and u - v must be in W.
- For any scalar k and u in W then k.u in W.
- 0 must be in W.
- For all u, v in W and u.v must be in W.

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MTH501 - Linear Algebra

MTH501:Grand Quiz

Quiz Start Time: 02:12 P

Question # 7 of 30 (Start time: 02:15:27 PM, 30 June 2021)

$$If \begin{pmatrix} 2x & 8y+4x \\ 8 & -7 \end{pmatrix} = \begin{pmatrix} 16 & 0 \\ 8 & -7 \end{pmatrix}, \ then \ (x,y) = ---.$$

Select the correct option

Reload M

- (-8,4)
- (-4,8)
- (8,-4)
- (4,-8)

Click to Save Answer & Move to N

MTH501:Grand Quiz

Question # 8 of 30 (Start time: 02:15:52 PM, 30 June 2021)

Which of the following is an example of Matrix in Echelon form?

Select the correct option

- $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$
- $\begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix}$
- $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$
- $\begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix}$

Click to Save

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MTH501 - Linear Algebra

MTH501:Grand Quiz

Quiz Start Time: 02

Question # 9 of 30 (Start time: 02:16:16 PM, 30 June 2021)

Let V be a vector space, and let W be a subset of V, what does it mean when we say that W is closed under addition?

Select the correct option

Relo

- ○ Whenever x,and y are in W, then $x + y$ is in V,
- ○ $W(x + y) = Wx + Wy$ for all two vectors x and y.
- ○ Every vector in W is the sum of two vectors in W,
- ● Whenever x and y are in W, then $x + y$ is in W,

Click to Save Answer & Move

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MTH501:Grand Quiz

Question # 10 of 30 (Start time: 02:16:42 PM, 30 June 2021)

If det(A) = 187 then the Cramer's rule can be applicable because:

Select the correct option

- ○ The coefficient matrix A is not invertible
- ○ The coefficient matrix A is m x n
- ● The coefficient matrix A is invertible
- ○ The coefficient matrix A is scalar

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MTH501:Grand Quiz

Question # 11 of 30 (Start time: 02:17:10 PM, 30 June 2021)

Gauss-Seidel method is also termed as a method of

Select the correct option

- False Position Method
- Elimination Method
- Iteration Method
- Successive Displacement

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MTH501:Grand Quiz

Quiz Start Time:

Question # 12 of 30 (Start time: 02:18:00 PM, 30 June 2021)

Suppose k is any scalar and $u = (u_1, u_2, \ldots, u_n)$, $v = (v_1, v_2, \ldots, v_n) \in R^n$, then the distributive law states that

Select the correct option

- K(u+v)=ku+kv
- k(u+v)=kuv

Click to Save Answer & Mo

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MTH501:Grand Quiz

Question # 13 of 30 (Start time: 02:18:36 PM, 30 June 2021)

Let A, B and C matrices are of same order and $A(B+C) = AB + AC$, this law is known as

Select the correct option

- ○ Associative law
- ● Distributive law
- ○ Commutative law
- ○ Cramer's law

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MTH501:Grand Quiz

Question # 14 of 30 (Start time: 02:19:03 PM, 30 June 2021)

If two rows or columns of a square matrix are identical, then det (A)wil be _____.

Select the correct option

- non zero
- zero
- positive
- one

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MTH501:Grand Quiz

Quiz Star

Question # 15 of 30 (Start time: 02:19:24 PM, 30 June 2021)

If a multiple of one row of a square matrix A is added to another row to produce a matrix B, then which of the following condition is true?

Select the correct option

- detA detB = detA
- detB = detA
- detB = k detA
- detA detB = 0

Click to Save Answe

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MTH501:Grand Quiz

Question # 16 of 30 (Start time: 02:19:51 PM, 30 June 2021)

The following Equations:

$$2x - y = 3$$
$$-x + \frac{1}{2}y = \frac{-3}{2},$$

is an example of - - - - - System.

Select the correct option

- Non-Linear
- Homogeneous
- Inconsistent
- Consistent

Click to Sav

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MTH501:Grand Quiz

Quiz Start T

Question # 17 of 30 (Start time: 02:20:18 PM, 30 June 2021)

Algebra is the study of the properties of operations on ------

Select the correct option

- Numbers
- Variables

Click to Save Answer

MTH501:Grand Quiz

Quiz Start Time: 02:12 PM, 30 June 2021

Question # 18 of 30 (Start time: 02:20:40 PM, 30 June 2021)

Total Marks: 1

If a system of equations is solved using the Gauss-Seidel method, then which of the following is the most appropriate answer about the matrix M that is derived from the coefficient matrix ?

Select the correct option

- All of its entries below and above the diagonal must be zero.
- All of its entries on the diagonal must be zero.
- All of its entries above the diagonal must be zero.
- All of its entries below the diagonal must be zero.

Click to Save Answer & Move to Next Question

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MTH501:Grand Quiz

Question # 19 of 30 (Start time: 02:21:06 PM, 30 June 2021)

What is the maximum possible number of pivots in a 4x6 matrix ?

Select the correct option

- 6
- 8
- 4
- 10

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MTH501:Grand Quiz

Question # 20 of 30 (Start time: 02:21:31 PM, 30 June 2021)

$If\ A = \begin{pmatrix} a & b \\ c & d \end{pmatrix},\ then\ which\ of\ the\ following\ is\ \ |A|$

Select the correct option

- bc+ad
- ad-bc
- ad+bc
- bc-ad

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MTH501:Grand Quiz

Question # 21 of 30 (Start time: 02:21:52 PM, 30 June 2021)

Basis is a spanning set that is as small as possible

Select the correct option

- (●) True
- ( ) False

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MTH501:Grand Quiz Quiz St

Question # 22 of 30 (Start time: 02:22:12 PM, 30 June 2021)

The equation : $0x - 0y = -5$ has - - - - solution(s).

Select the correct option

- Infinite many
- Unique
- No
- distinct finite

Click to Save Answ

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Quiz Start

Question # 23 of 30 (Start time: 02:22:35 PM, 30 June 2021)

$$\begin{vmatrix} \int_0^1 e^x\,dx & \int_0^1 e^{1-x}\,dx \\ -\int_0^1 e^{1-x}\,dx & \int_0^1 e^x\,dx \end{vmatrix} = - - - -$$

Select the correct option

- $e-1$
- $(e-1)^2$
- $2(e-1)^2$
- 0

Click to Save Answer

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Question # 24 of 30 (Start time: 02:23:02 PM, 30 June 2021)

$3x + 2y - 3$ is an algebraic

Select the correct option

- ○ Equation
- ○ Sentence
- ○ Inequality
- ● Expression

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MTH501:Grand Quiz

Question # 25 of 30 (Start time: 02:23:32 PM, 30 June 2021)

All the lines those passes through origin are not the subspace of a plane.

Select the correct option

- TRUE
- FALSE

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Quiz Start Ti

Question # 26 of 30 (Start time: 02:23:58 PM, 30 June 2021)

A linear equation in three variables always represent a - - - - - .

Select the correct option

- Line in $\mathbb{R}^2(2-dimension)$
- Line in$\mathbb{R}^3(3-dimension)$
- Plane in$\mathbb{R}^3(3-dimension)$
- Plane in$\mathbb{R}^2(2-dimension)$

Click to Save Answer &

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Question # 27 of 30 (Start time: 02:24:32 PM, 30 June 2021)

The Invertible Matrix Theorem applies only to _____ matrices.

Select the correct option

- identity
- rectangular
- square
- scalar

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Quiz Start

Question # 28 of 30 (Start time: 02:24:57 PM, 30 June 2021)

If $\alpha = \beta$, then $\begin{vmatrix} Cos\alpha & -Sin\alpha \\ Cos\beta & Sin\beta \end{vmatrix} =$

Select the correct option

- 1
- Undefined
- (∞)
- $Sin2\alpha$
- $Cos2\alpha$

Click to Save Answer

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Question # 29 of 30 (Start time: 02:25:33 PM, 30 June 2021)

A partitioned matrix is a partition of a matrix into rectangular smaller matrices called _____.

Select the correct option

- row
- blocks
- column
- element

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MTH501 - Linear Algebra

MTH501:Grand Quiz

Quiz Start Time: 02:

Question # 30 of 30 (Start time: 02:25:58 PM, 30 June 2021)

As soon as an approximate solution of a linear system is sufficiently accurate for practical work then which of the following is true for an iterative process?

Select the correct option

- The process may be carried on.
- The reduced row-echelon method may be used.
- The row-echelon method may be used.
- The process may be stopped.

Click to Save Answer & Move

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Quiz Start Time: 02:

Question # 30 of 30 (Start time: 02:25:58 PM, 30 June 2021)

As soon as an approximate solution of a linear system is sufficiently accurate for practical work then which of the following is true for an iterative process?

Select the correct option

- The process may be carried on.
- The reduced row-echelon method may be used.
- The row-echelon method may be used.
- The process may be stopped.

Click to Save Answer & Move

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MTH501:Grand Quiz

Question # 1 of 30 (Start time: 02:28:15 PM, 30 June 2021)

For a non-singular matrix A, A^{-1} is equal to

Select the correct option

- $det(A) * A$
- $det(A) * Adj(A)$
- None of the above.
- $Adj(A)/det(A)$

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Question # 2 of 30 (Start time: 02:28:52 PM, 30 June 2021)

If a matrix is in reduced row echelon form, then it is also in row echelon form.

Select the correct option

- True
- May be
- None of the above
- False

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MTH501:Grand Quiz

Question # 3 of 30 (Start time: 02:29:13 PM, 30 June 2021)

Consider an LU-factorization, the matrix L is _____ and is called a unit lower triangular matrix.

Select the correct option

- identity
- invertible
- zero
- singular

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MTH501:Grand Quiz

Question # 4 of 30 (Start time: 02:29:40 PM, 30 June 2021)

The determinant of the matrix A is denoted by

Select the correct option

- $\bar{A}$
- $|A|$
- A^T
- A'

Cli

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MTH501 - Linear Algebra

MTH501:Grand Quiz

Quiz Start Time: 02:28 PM, 30 June 2021

Question # 5 of 30 (Start time: 02:30:04 PM, 30 June 2021)

Total Marks: 1

If the coefficient matrix is large and with a high proportion of zero entries, then which of the following works quickly in favor of solving a linear systems for saving less computer memory?

Select the correct option

- Row-echelon method
- Direct method
- Iterative method
- Reduced row-echelon method

Click to Save Answer & Move to Next Question

MTH501:Grand Quiz

Quiz

Question # 6 of 30 (Start time: 02:30:37 PM, 30 June 2021)

$$If\ \alpha = \beta,\ then\ \begin{vmatrix} Cos\alpha & Sin\alpha \\ Sin\beta & Cos\beta \end{vmatrix} = - - - - - -.$$

Select the correct option

- ∞
- 1
- $Cos2\alpha$
- $Sin2\alpha$

Click to Save A

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MTH501:Grand Quiz

Quiz Start Time: 02:

Question # 7 of 30 (Start time: 02:31:11 PM, 30 June 2021)

Let V be a vector space, and let W be a subset of V, what does it mean when we say that W is closed under addition?

Select the correct option

Reloa

- Whenever x,and y are in W, then $x + y$ is in V,
- Every vector in W is the sum of two vectors in W,
- Whenever x and y are in W, then $x + y$ is in W,
- $W(x + y) = Wx + Wy$ for all two vectors x and y.

Click to Save Answer & Move

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MTH501:Grand Quiz

Question # 8 of 30 (Start time: 02:31:37 PM, 30 June 2021)

A system of linear equations is said to be homogeneous if the constant terms are all

Select the correct option

- (●) Zero
- ( ) Both (a) and (b)
- ( ) One
- ( ) None of the above

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MTH501:Grand Quiz Quiz

Question # 9 of 30 (Start time: 02:32:01 PM, 30 June 2021)

If M is a square matrix having two rows equal then which of the following about the determinant of the matrix is true?

Select the correct option

- det (M)=0
- det (M)=1
- det (M) is not equal to '1'
- det (M) is not equal to '0'

Click to Save An

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MTH501:Grand Quiz

Question # 10 of 30 (Start time: 02:32:23 PM, 30 June 2021)

What is the maximum possible number of pivots in a '3 by 3' matrix ?

Select the correct option

- 3
- 5
- 0
- 1

Click to S

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MTH501:Grand Quiz

Quiz Start Time

Question # 11 of 30 (Start time: 02:32:49 PM, 30 June 2021)

Which of the following is the area of the parallelogram determined by the columns of A where A is a 2 x 2 matrix?

Select the correct option

- ● |det A|
- ○ [A]
- ○ det A
- ○ A^(-1) ,that is inverse of A

Click to Save Answer & M

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MTH501:Grand Quiz

Question # 12 of 30 (Start time: 02:33:22 PM, 30 June 2021)

Any set of linearly independent vectors cannot contain the ---------- vector.

Select the correct option

- None of these
- Unit
- zero
- Identity

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MTH501:Grand Quiz

Quiz Start Tin

Question # 13 of 30 (Start time: 02:33:47 PM, 30 June 2021)

A matrix can be subdivided into sub matrices in various ways by inserting lines between selected _____ and columns.

Select the correct option

- blocks
- diagonals
- rows
- elements

Click to Save Answer &

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MTH501 - Linear Algebra

MTH501:Grand Quiz

Quiz Start Time: 02:28 PM, 30 Jur

Question # 14 of 30 (Start time: 02:34:24 PM, 30 June 2021)

Total M

Let A be the matrix of order 2X3 and B be the matrix of order 3X5,then which of the following is the order of the matrix AB?

Select the correct option

- 2X3
- 3X5
- 3X3
- 2x5

Click to Save Answer & Move to Next Ques

MTH501:Grand Quiz

Quiz Start Time: 02:28 PM,

Question # 15 of 30 (Start time: 02:34:55 PM, 30 June 2021)

If $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \underset{\sim}{\text{Row Equivalent}} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, then the Columns of A are Linearly - - - - - - -

Select the correct option

Reload Mat

- Independent
- Dependent

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MTH501:Grand Quiz

Question # 16 of 30 (Start time: 02:35:20 PM, 30 June 2021)

$3x + 2y - 3$ is an algebraic

Select the correct option

- ● Expression
- ○ Sentence
- ○ Equation
- ○ Inequality

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MTH501:Grand Quiz

Question # 17 of 30 (Start time: 02:35:44 PM, 30 June 2021)

Any system of n linear equations in (n+1) variable can be solved by Cramer's rule.

Select the correct option

- ○ TRUE
- ● FALSE

Cli

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MTH501:Grand Quiz

Question # 18 of 30 (Start time: 02:36:14 PM, 30 June 2021)

If one row of A is multiplied by k to produce B, then which of the following condition is true?

Select the correct option

- detB = - detA
- detB = detA
- det(AB) = (detA)(detB)
- detB = k detA

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MTH501:Grand Quiz

Question # 19 of 30 (Start time: 02:36:41 PM, 30 June 2021)

Let W = {(x, y) such that x, y in R and x = y}. Is W a vector subspace of plane.

Select the correct option

- NO
- YES

Click to Sav

MTH501:Grand Quiz

Quiz Start Time: 02:28 PM, 30 June 2

Question # 20 of 30 (Start time: 02:37:04 PM, 30 June 2021)

Total Marl

A transformation $T : R^n \rightarrow R^m$ Is a rule that assigns to each vector x in R^n , an image vector T(x) in R^m. The set R^n is called the_____ , and R^m is called the _____.

Select the correct option

Reload Math Equatio

- co-domain of T, domain of T
- domain of T, co-domain of T

Click to Save Answer & Move to Next Questio

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MTH501:Grand Quiz

Question # 21 of 30 (Start time: 02:37:31 PM, 30 June 2021)

For any 3x3 matrix A where det (A) = 3, then det (2A) = _____.

Select the correct option

- 20
- 15
- 6
- 24 (selected)

Click to

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MTH501:Grand Quiz

Question # 22 of 30 (Start time: 02:37:58 PM, 30 June 2021)

According to determinant properties, when two rows are interchanged then the sign of determinant

Select the correct option

- ● Must change
- ○ Remains same
- ○ Multiplied
- ○ Divided

Click to Save

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MTH501:Grand Quiz

Question # 23 of 30 (Start time: 02:38:23 PM, 30 June 2021)

The order of matrix [1 3 7] is

Select the correct option

- 2-by-2
- 1-by-3 (●)
- 2-by-1
- 1-by-2

Click to

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MTH501:Grand Quiz

Question # 24 of 30 (Start time: 02:38:46 PM, 30 June 2021)

For which of the matrix, the Gauss-Seidel method is applicable?

Select the correct option

- Strictly diagonally dominant matrix
- Null matrix
- Diagonally dominant matrix
- Rectangular matrix

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MTH501:Grand Quiz

Question # 25 of 30 (Start time: 02:39:05 PM, 30 June 2021)

According to determinant properties, the determinants of resulting matrix equal to k delta if elements of rows are

Select the correct option

- Divided to constant k
- Multiplied to constant k
- Added to constant k
- Subtracting to constant k

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MTH501:Grand Quiz Q

Question # 26 of 30 (Start time: 02:39:35 PM, 30 June 2021)

The Equations: $\begin{matrix} 2x - y = 6 \\ -x + \frac{1}{2}y = \frac{-3}{2}, \end{matrix}$

is an example of - - - - - System.

Select the correct option

- Non-Linear
- Homogeneous
- Consistent
- Inconsistent

Click to Save

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MTH501 - Linear Algebra

MTH501:Grand Quiz

Quiz S

Question # 27 of 30 (Start time: 02:39:59 PM, 30 June 2021)

For the following system of Equations:
$x + 2y = 0, 2x - y = 0;$
which variable can be taken as free?

Select the correct option

- both x and y
- only y
- either x or y
- only x

MTH501:Grand Quiz

Quiz Start Time: 02:28

Question # 28 of 30 (Start time: 02:40:24 PM, 30 June 2021)

Which of the following would be the Augmented Matrix associated with the system: $\begin{matrix} 2x = 1 \\ -2y = x? \end{matrix}$

Select the correct option

Reload

- $\left(\begin{array}{cc|c} 2 & 0 & 1 \\ -1 & -2 & 1 \end{array}\right)$
- $\left(\begin{array}{cc|c} 2 & 0 & 1 \\ -1 & -2 & 0 \end{array}\right)$
- $\left(\begin{array}{cc} 2 & 1 \\ -2 & 1 \end{array}\right)$
- $\left(\begin{array}{cc|c} 2 & 0 & 1 \\ 1 & -2 & 0 \end{array}\right)$

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Quiz

Question # 29 of 30 (Start time: 02:40:53 PM, 30 June 2021)

A linear equation in three variables always represent a - - - - - .

Select the correct option

- ● Plane in $\mathbb{R}^3(3-dimension)$
- ○ Line in $\mathbb{R}^3(3-dimension)$
- ○ Plane in $\mathbb{R}^2(2-dimension)$
- ○ Line in $\mathbb{R}^2(2-dimension)$

Click to Save An

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MTH501:Grand Quiz

Question # 30 of 30 (Start time: 02:41:18 PM, 30 June 2021)

The computational efficiency of the LU factorization depends on knowing _____ matrix (matrices).

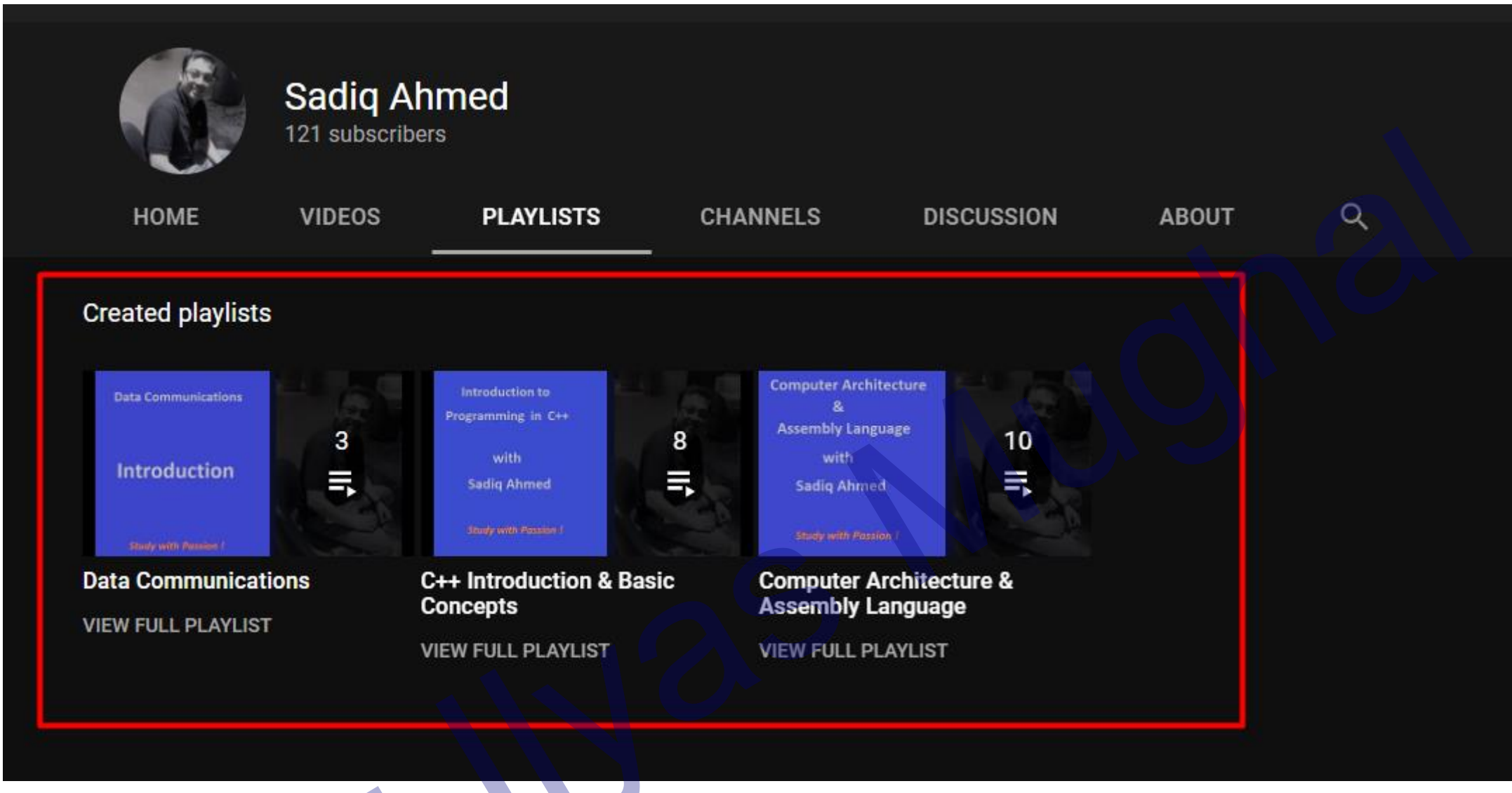
Select the correct option

- (c) diagonal
- (a) lower triangular
- (d) both (a) and (b)
- (b) upper triangular

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https://www.youtube.com/channel/UCjboEX65C_yN7as5di1OuIQ/playlists

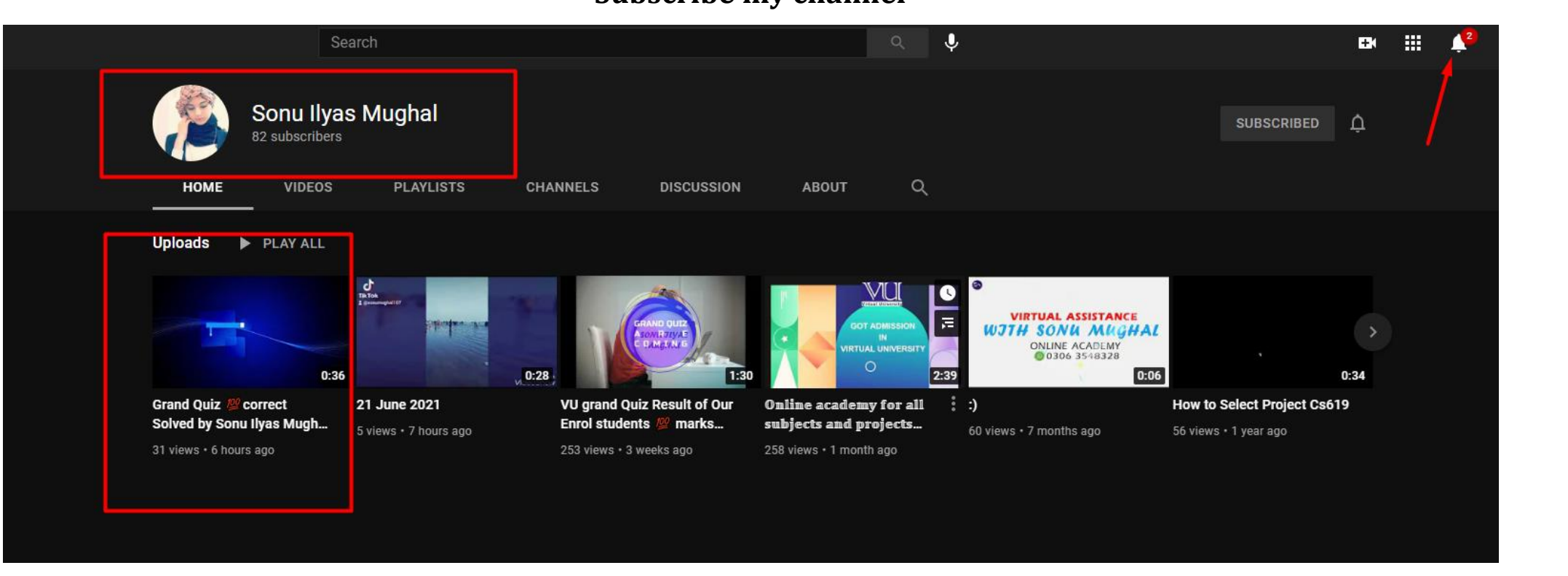


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c) Final report
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